

Measuring AI Ability to Complete Long Software Tasks

Time Horizon Doubling Every 7 Months

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A New Metric: 50% Task-Completion Time Horizon

Step 1

Assemble diverse task suite

170 software/research tasks ranging from 1 sec to 8 hrs.

Step 2

Measure baseline & AI success

Time skilled humans, evaluate AI agent success rates.

Step 3

Fit logistic model

Estimate task duration at which AI achieves 50% success.

Family	Length	Description
find_shell_script	3 seconds	Multiple choice: “Which file is a shell script?” Choices: “run.sh”, “run.txt”, “run.py”, “run.md”
wikipedia_research	1 minute	Research simple factual information from Wikipedia and provide accurate answers to straightforward questions.
oxdna_simple	9 minutes	Detect and fix a bug in the input files for a molecular dynamics simulation using the oxDNA package.
munge_data	56 minutes	Write a Python script to transform JSON data from one format to another by inferring the conversion rules from provided example files.
cuda_backtesting	8 hours	Speed up a Python backtesting tool for trade executions by implementing custom CUDA kernels while preserving all functionality, aiming for a 30x performance improvement.

Methodology Overview

170 Tasks Spanning 3 Seconds to 8 Hours of Human Effort

Task Suite Composition

Suite	Tasks	Duration Range	Description
HCAST	97	1 min - 30 hrs	Diverse software tasks
RE-Bench	7	8 hrs	Difficult ML research engineering tasks
SWAA	66	1 sec - 30 sec	Software Atomic Actions (single-step)

Example Tasks

Family	Human Time	Description
find_shell_script	3 seconds	Multiple choice: "Which file is a shell script?"
wikipedia_research	1 minute	Research simple factual information from Wikipedia
oxdna_simple	9 minutes	Detect and fix a bug in molecular dynamics simulation input files
munge_data	56 minutes	Write a Python script to transform JSON data by inferring rules
cuda_backtesting	8 hours	Implement custom CUDA kernels for 30x speedup of backtesting tool

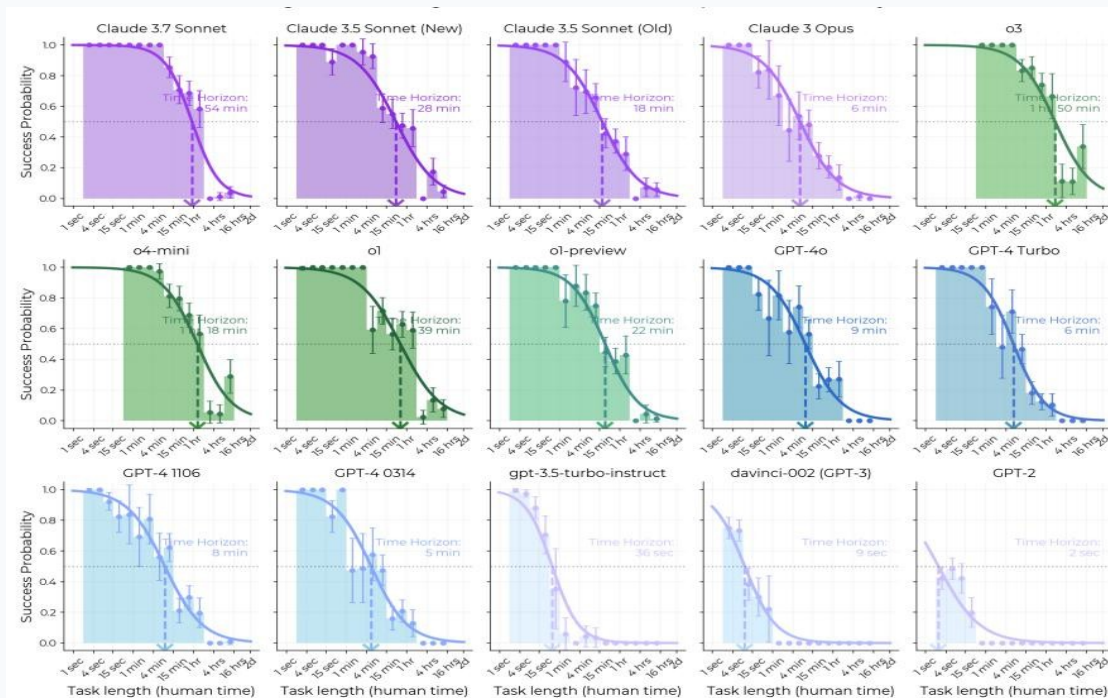
AI Task Horizon Grew from 2 Secs to ~1.8 Hours in 6 Years

Key Milestones

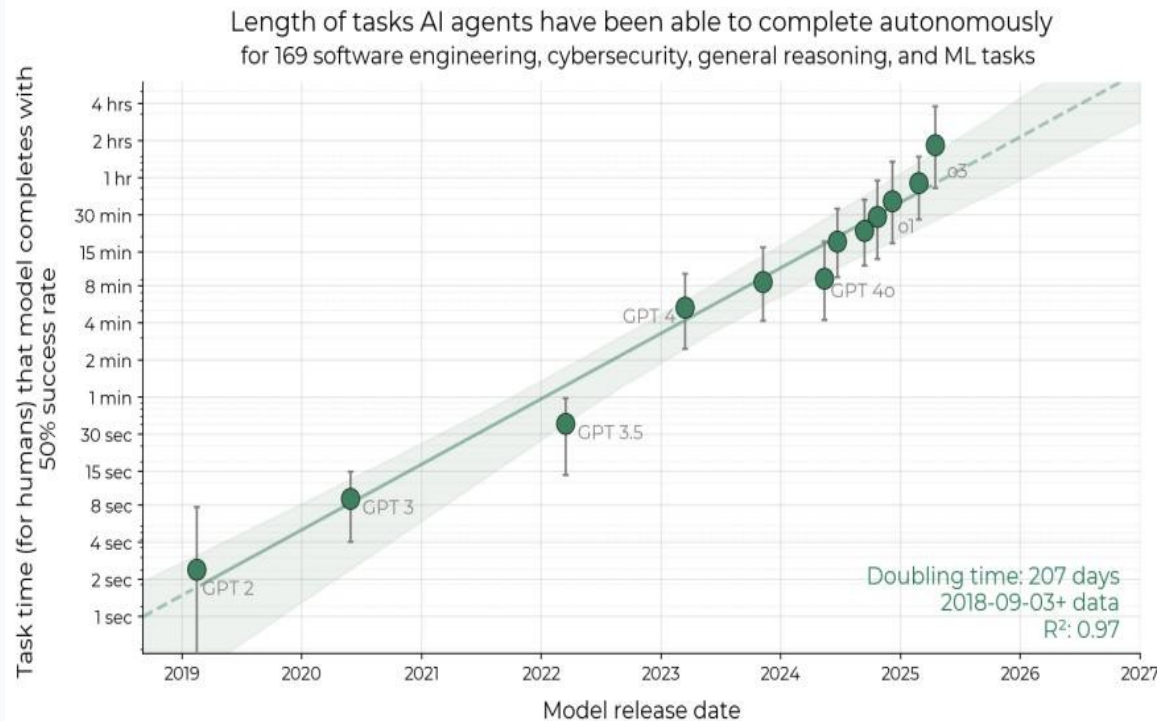
- GPT-2 (2019): ~2 seconds
- GPT-4 (2023): ~5 minutes
- Claude 3.5 Sonnet: ~28 minutes
- o3 (2025): ~1 hour 50 minutes

Growth Trend

- Doubling time: ~207 days (~7 mos)
- Excellent fit: $R^2 = 0.97$
- 2023-2025 rate ~20% faster than overall



Success Probability Drops Sharply as Task Length Increases



Rightward Shift

Frontier models are significantly pushing the success curve rightward.

Claude 3.7 Sonnet (54m) and o3 (1h50m) demonstrate robust success on complex tasks compared to older models like GPT-2 (2s).

Reasoning, Tool Use, and Error Recovery Drive Growth

Improved Logical Reasoning

Models possess deeper capacity to break down complex tasks and plan sequentially.

Advanced Tool Use

Better utilization of environments, terminals, and external APIs directly within execution loops.

Error Recovery & Adaptability

Greater reliability and self-awareness allowing models to detect mistakes and self-correct during long trajectories.

Limitation: Unstructured Tasks

Performance remains notably lower on "messier", less structured tasks where evaluation criteria are subjective or highly open-ended.

Extrapolation Predicts Month-Long Automation by 2028-2031

A 1-Month Time Horizon (167 Work Hours) by Mid-2028 to Mid-2031

Trend Acceleration

The 2023–2025 growth rate is approximately 20% faster than the overall 2019–2025 rate, suggesting the possibility of accelerated progress rather than plateauing in the near term.

Caveats & Open Questions

- **External Validity:** The tasks do not perfectly represent the average segment of real-world intellectual labor. It's uncertain how well this generalizes to complex corporate jobs.
- **Trend Stability:** Extrapolating exponential trends always carries risk, hitting a "wall" in scaling or

Takeaways: Capability Growth Demands Proactive Governance

1

Robust Metric: 50% time horizon is a reliable, intuitive metric for tracking AI autonomy ($R^2=0.97$ over 6 years).

2

Relentless Pace: Doubling every ~7 months with possible acceleration since 2024.

3

Current Frontier: o3 can autonomously complete tasks that take skilled humans ~2 hours.

4

Uncertain Scope: External validity remains an open question—tasks may not perfectly represent real-world distributions.

5

Current Limits: Models still struggle with poor performance on unstructured, "messy" tasks.